COMPUTER ENGINEERING

QUEZON CITY

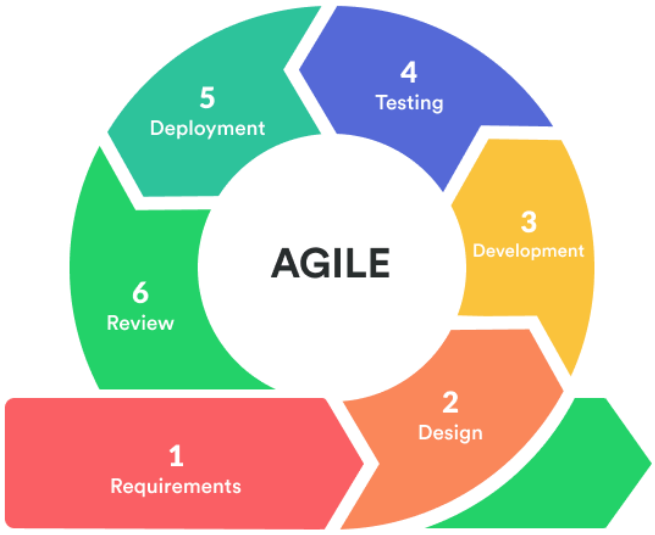
COURSE CODE/TITLE:	CPE010 / DATA STRUCTURE AND ALGORITHM	**SECTION:**	CPE21S4
DATE PERFORMED:	Oct 9, 2026	**DATE SUBMITTED:**	Oct 9, 2026
NAME:	GROUP 7 Lacsamana, Timothy Justin Macaraeg, Bien Glen	**INSTRUCTOR:**	Engr. Jimlord Quejado

Seatwork 11.1 **Planning**

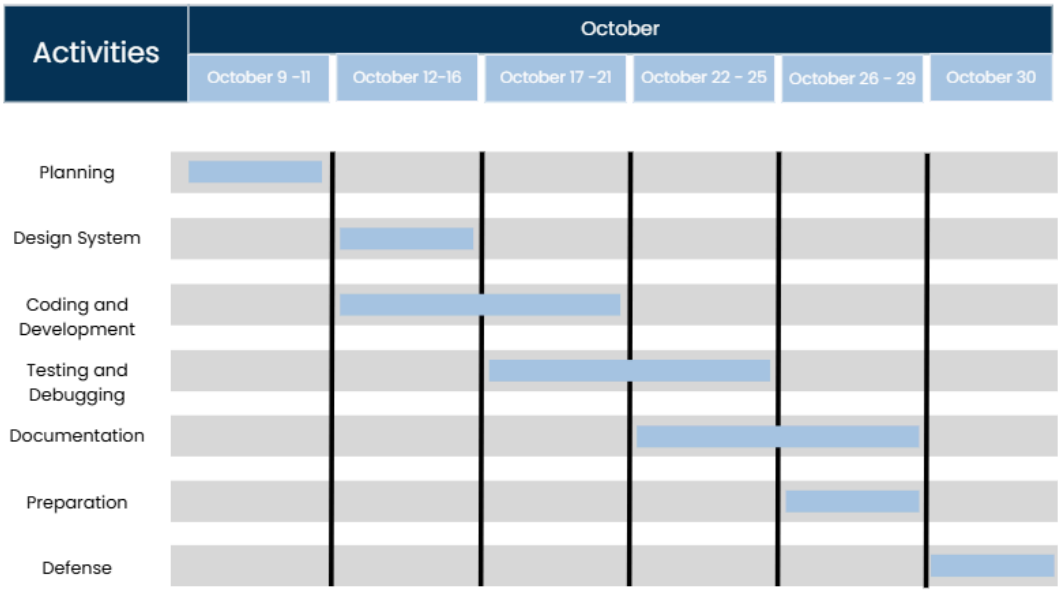
1. PROCEDURE OUTPUT

Billiards Hall Management and Table Reservation System	This system will help billiards hall staff and owners to manage table availability, customer reservations, playing time, and payments. It will also allow them to monitor customer queues, calculate playing fees, and keep records of daily sales and transactions.
Tables and Chairs Rental Inventory and Reservation System	This system will help rental business owners manage their tables and chairs inventory, customer bookings, and rental schedules. It will also help them record payments, and return dates, and keep track of damaged or missing items
Cosplay Costume and Equipment Rental Management System	This system will help cosplay rental businesses organize their costumes, accessories, and equipment. It will allow them to monitor item availability, manage customer reservations, record rental fees and payments, and track returned, damaged, or missing items.

Our group decided to use the **Agile Model** because it allows us to develop and improve our proposed systems step by step. This approach can be applied to all three proposed systems: the Billiards Hall Management and Table Reservation System, Tables and Chairs Rental Inventory and Reservation System, and Cosplay Costume and Equipment Rental Management System. We can divide the development process into smaller tasks, test each feature, and make changes based on feedback from our group and potential users.



Gantt Chart for DSA System



Task	Responsibility
Planning	Discuss the proposed systems, choose the system to develop, and identify its objectives and requirements.
System Design	Create the system flowchart, plan the program structure, and design the user interface.
Coding and Development	Write the C++ code and implement the system features using appropriate data structures and algorithms.
Testing and Debugging	Test the system, find errors, check each feature, and fix problems in the code.
Documentation	Prepare the project documentation, explain the DSA concepts used, and organize the screenshots and results.
Preparation	Prepare the presentation, divide the speaking parts, and practice explaining and demonstrating the system.
Defense	Present the completed system, demonstrate its features, and answer the questions from the instructor.

10 Questions

Billiards Hall Management and Table Reservation System

1. How do you keep track of which billiards tables are available or currently occupied?
2. How do you record the time when a customer starts and finishes playing?
3. How do you calculate the total fee based on the customer's playing time?
4. How do you handle customers who are waiting when all tables are occupied?
5. Do you accept advance reservations, and how do you record them?
6. How do you record customer payments and calculate the total daily sales?
7. What do you do when a customer wants to extend their playing time?
8. How do you check previous transactions if a customer has a concern about their payment?
9. How do you manage tables that need cleaning, maintenance, or repair?
10. What common problems do you experience when managing the billiards hall, and what features would you like to have in a computerized system?

Tables and Chairs Rental Inventory and Reservation System

1. How do you keep track of the available tables and chairs for rent?
2. How do you record customer reservations and rental schedules?
3. How do you calculate the total rental fee based on the number of items and rental duration?
4. How do you handle customers who want to change their reservation date?
5. How do you record customer information and contact details?
6. How do you monitor the delivery and pickup of rented tables and chairs?
7. What do you do if some tables or chairs are damaged or missing after being returned?
8. How do you record customer payments and remaining balances?
9. How do you check which items are available for a specific date?
10. What common problems do you experience in managing your rental business, and what features would you like in a computerized system?

Cosplay Costume and Equipment Rental Management System

1. How do you keep track of the available cosplay costumes and equipment for rent?
2. How do you record customer reservations and the scheduled rental dates?
3. How do you calculate the rental fee based on the costume or equipment and rental duration?
4. How do you handle customers who want to extend their rental period?
5. How do you record customer information and contact details?
6. How do you check if a costume or equipment is available for a specific date?
7. What do you do if a costume or piece of equipment is damaged or lost?
8. How do you record payments, deposits, and additional charges?
9. How do you monitor the return dates of rented costumes and equipment?
10. What common problems do you experience in managing cosplay rentals, and what features would you like in a computerized system?

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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